INCH-POUND
MIL-C-19836B(SH)
23 March 1988
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31 August 1960
(See 6.8)

MILITARY SPECIFICATION

COOLERS, FLUID, INDUSTRIAL, AIR, MOTOR AND GENERATOR, NAVAL SHIPBOARD

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers air coolers for motors, generators and similar electrical equipment for Naval shipboard application. These coolers commonly form part of the water-air-cooled enclosures built in conformance with MIL-E-2036.

1.2 Classification. The coolers shall be of the following classes as specified (see 6.2.1):

Class 1 - Submarine, salt water cooled, salt water side subject to sea pressure for greater than 200 feet submergence, cyclic life stipulated.
Class 2 - Submarine, salt water cooled, salt water side secured for greater than 200 feet submergence.
Class 3 - Surface ship, salt water cooled.
Class 4 - Submarine and surface ship fresh water cooled.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specificatons and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-B-639	- Brass, Naval: Flat Products (Plate, Bar, Sheet, and Strip).
QQ-B-750	- Bronze, Phosphor; Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.
QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
QQ-C-465	- Copper-Aluminum Alloys (Aluminum Bronze) (Copper Alloy Numbers 606, 614, 630, 632M, and 642); Rod, Flat Products with Finished Edges (Flat Wire, Strip, and Bar), Shapes, and Forgings).
QQ-C-591	- Copper-Silicon, Copper-Zinc-Silicon, and Copper-Nickel-Silicon Alloys: Rod, Wire, Shapes, Forgings, and Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate).
QQ-N-281	- Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286	- Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-P-1757	- Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.
PPP-B-591	- Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock-Corner.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.

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MIL-C-104	- Crate, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-A-907	- Antiseize Thread Compound, High Temperature.
MIL-G-1149	- Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness.
MIL-S-1222	- Studs, Bolts, Hex Cap Screws, Socket Head Cap Screws and Nuts.
MIL-C-3774	- Crates, Wood; Open 12,000-and 16,000-Pound Capacity.
MIL-G-5514	- Gland Design; Packings, Hydraulic, General Requirements for.
MIL-R-6855	- Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for.
MIL-S-8660	- Silicone Compound, NATO Code Number S-736.
MIL-T-15005	- Tubes, 70-30 and 90-10 Copper Nickel Alloy, Condenser and Heat Exchanger.

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MILITARY (Continued)

MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
DOD-P-15328 - Primer (Wash), Pretreatment (Formula No. 117 for Metals). (Metric)
MIL-C-15726 - Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.
MIL-E-15809 - Expander, Tube, Condenser and Heat Exchangers.
MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706).
MIL-A-18001 - Anodes, Corrosion Preventive, Zinc; Slab Disc and Rod Shaped.
MIL-A-19521 - Anodes, Corrosion Preventive, Zinc, and Plugs, Zinc Anode Retaining; Design of and Installation in Shipboard Condensers and Heat Exchangers.
MIL-C-20159 - Copper-Nickel Alloy Castings (UNS No. C96200 and C96400).
MIL-G-21610 - Gaskets, Heat Exchanger, Various Cross Section Rings, Synthetic Rubber.
MIL-S-22473 - Sealing, Locking, and Retaining Compounds: Single-Component).
MIL-S-22698 - Steel Plate and Shapes, Weldable Ordinary Strength and Higher Strength: Hull Structural.
MIL-N-24106 - Nickel-Copper Alloy Bars, Rods, and Forgings.
MIL-T-24107 - Tube, Copper (Seamless) (Copper Alloy Numbers C10100, C10200, C10300, C10800, C12000, C12200, and C14200).
MIL-L-24131 - Lubricant, Colloidal Graphite in Isopropanol.
MIL-L-24479 - Lubricant, Red Lead and Graphite in Mineral Oil.
MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service.
DOD-G-24508 - Grease, High Performance, Multi-Purpose. (Metric)
MIL-C-52950 - Crate, Wood, Open and Covered.
MIL-R-83248 - Rubber, Fluorocarbon Elastomer, High Temperature, Fluid, and Compression Set Resistant.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-22 - Welded Joint Designs.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-438 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Submarines Service.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
DOD-STD-1399, Section 301 - Interface Standard for Shipboard Systems, Ship Motion and Attitude. (Metric)

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MS16142 - Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions for.
MS17828 - Nut, Self-Locking, Hexagon, Regular-Height, (Nonmetallic Insert) 250°F, Nickel-Copper Alloy.
MS17829 - Nut, Self-Locking, Hexagon, Regular Height, 250°F, (Non-metallic Insert) Non-Corrosion-Resistant Steel.
MS18229 - Plug for "O" Ring Gasket.

2.1.2 Other Government drawing and publication. The following other Government drawing and publication form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
803-5959186 - Submarine Heat Exchanger Anode Plug.

PUBLICATION

NAVSEA
0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
B1.12 - Class 5 Interference-Fit Thread.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
ASME Boiler and Pressure Vessel Code
Section III - Rules for Construction of Nuclear Power Plant Components.
Section VIII, Division I - Pressure Vessels.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 106 - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service. (DoD adopted)

B 171 - Standard Specification for Copper-Alloy Plate and Sheet for Pressure Vessels, Condensers, and Heat Exchangers.

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Structural requirements. The structural design calculations for class 1 coolers shall be in accordance with Section III (class 1 vessel) of ASME Boiler and Pressure Vessel Code. Structural design calculations for coolers other than class 1 shall be in accordance with ASME Boiler and Pressure Vessel Code, Section VIII, Division I.

3.1.2 Space and weight. Space occupied and weight of coolers shall be held to a minimum consistent with meeting the requirements of this specification and stipulated performance requirements.

3.1.2.1 Submarine installations. The dimensions of air coolers to be installed in submarines shall be such that the coolers can be removed through a standard hatch, or in case there is no hatch in the compartment in which the unit is to be installed, such that the unit can be moved through bulkhead openings into a compartment provided with a hatch. Hatch and bulkhead opening dimensions will be as specified (see 6.2.1).

3.1.3 Requirements due to ship attitude or motion. Unless otherwise specified (see 6.2.1), coolers shall meet the requirements of DOD-STD-1399, section 301 in regards to ship motion and attitude.

3.1.4 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.1.5 Mercury exclusion. Air coolers shall not contain mercury in any form. Mercury-bearing instruments or equipment which might cause contamination shall not be used in the manufacture, fabrication, assembly or testing of any

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material. The most probable causes of contamination are directly connected manometers, mercury vacuum pumps, mercury seals, mercury-in-glass thermometers, or handling of mercury in the immediate vicinity. In case of doubt or question of manufacturing procedures, contact NAVSEA or the agency concerned for assistance. In the event of any accident involving mercury contamination of the material being furnished or suspicion of such contamination, NAVSEA or the agency concerned shall be notified immediately.

3.1.6 Welding and allied processes. Fabrication welding and inspection shall be in accordance with MIL-STD-278, except as modified herein. For class 1 coolers, welding on parts subjected to sea pressure shall be in accordance with class A-F of MIL-STD-278, except as modified herein. Welded joints on the salt water side of class 1 coolers shall be radiographable, except for vent and drain nipples. Waterbox vent and drain nipple welded connections of class 1 coolers shall meet the 100 percent weld efficiency requirement for pressure vessels.

3.1.7 Casting inspection and repair. Casting inspection and repair shall be in accordance with MIL-STD-278.

3.1.8 Brazing. Brazing shall be in accordance with NAVSEA 0900-LP-001-7000, except that requirements for use of preinserted brazing rings are applicable only to pipe fittings. Brazed joints except where used in outer tube to inner tube sheet joints shall be ultrasonically inspected. Brazing shall not be used on the sea water side of submarine coolers (for example, class 1 or class 2 coolers).

3.1.9 Screw threads. Thread form, dimensions, and tolerances shall be in accordance with FED-STD-H28.

3.1.9.1 Threaded parts. Threaded parts shall conform to MIL-S-1222, except that materials shall be as specified in 3.2. The construction shall be such that standard wrenches can be used throughout. Tapered pipe threads shall not be used. The practice of bottoming or shouldering studs shall not be used. For the set end of studs, a class 3 fit used with locking compound in accordance with MIL-S-22473, grade AV, or a class 5 interference fit conforming to ANSI B1.12 shall be used. Threaded plugs, threaded connections, and threaded closures are prohibited in the submergence pressure boundary of class 1 coolers.

3.1.9.1.1 Thread lubricants. Threaded fasteners with torque requirements shall be lubricated before assembly. The lubricant used shall be identified on applicable drawings and it shall meet the following requirements as applicable:

(a) For class 1 coolers, the lubricant for bolting subjected to submergence pressure shall conform to MIL-L-24479.
(b) For coolers in saturated steam plant ships, lubricant shall be graphite in isopropanol and shall conform to MIL-L-24131 (Military symbol CGI).
(c) For all other bolting, the lubricant shall conform to MIL-A-907.

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3.1.9.1.2 Preferred fastener types. Preferred fastener types are listed in order of preference:

(a) Through bolt or through (2-nut) stud.
(b) Tap-end stud (one-nut).
(c) Cap screw or cap bolt.

3.1.9.1.3 Cap screws and cap bolts. Cap screws and cap bolts shall not be used for waterbox-to-shell or waterbox-to-tube sheet bolting, for inspection cover bolting, or for zinc anode support cover bolting.

3.1.9.1.4 Collar or stud bolts. Where collar bolts or stud bolts are used to make up a multiple flange joint, so that it can be selectively disassembled, such bolts or studs have a square extension beyond the threads on one end, for use of a wrench to prevent turning of the bolt when the nuts are tightened or removed.

3.1.9.1.5 Thread engagement. Stud engagement shall be sufficient to hold the maximum design load. In no case shall engagement be less than the nominal diameter of the fastener. Nuts shall be fully engaged in all cases. For a threaded fastener, not less than one thread but no more than four threads shall protrude beyond the crown of the nut.

3.1.9.1.6 Waterbox and tube sheet bolting. For all waterbox and tube sheet bolting, the following requirements shall apply:

(a) The bolting shall be tightened to a stipulated torque value by means of a torque wrench.
(b) For class 1 coolers, bolting construction including stipulated torque values shall be established and supporting calculations shall be furnished.
(c) For class 1 coolers, the bolting subjected to submergence pressure shall meet the following requirements:

(1) Calculated bolt stress due to design pressure shall be limited to 1/3 of the yield strength of the material at the design temperature.
(2) Maximum value of service stress averaged across bolt cross section and neglecting stress concentration factor shall be limited to 2/3 of the yield strength of the material of the design temperature.
(3) Maximum value of service stress at periphery of the bolt cross section resulting from tension, plus bending (neglecting stress concentration factor) shall be limited to the yield strength of the material at the design temperature.
(4) The value of installation torque specified on the drawings shall be calculated as follows:

a. Using P as the hydrostatic test load or load based upon design pressure plus maximum piping load on nozzles, whichever is greater, and high coefficient of friction ($\mu = 0.13$), determine torque using the following relation:

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$(P/T)\mu$ = Load per unit torque corresponding to bolt size and lubricant

$$(P/T)\mu = \frac{12}{r_t \left[\frac{\cos\theta \tan\alpha + \mu}{\cos\theta - \mu \tan\alpha} + \frac{r_c \mu}{r_t} \right]}$$

Where: r_t = minimum pitch radius of external threads (inches)

θ = 1/2 of thread angle (degrees)

α = Helix angle (degrees) equal arc $\tan \left[1/(2 \pi r_t N) \right]$

r_c = Collar radius, average of $D_h/2$ and bolt major radius (inches)

N = Number of threads per inch

μ = 0.13

T = Torque (foot-pounds)

D_h = Diameter across flats of nuts (average of maximum and minimum) (inches)

D = Average of mean pitch and minor diameters of external thread (inches) equals $2 r_t - (0.32476/N)$

b. Using the installing torque above and low coefficient of friction (μ = 0.065), determine stress from the following equation:

$$S_{av} = \frac{P}{6\pi\mu^2 D_h^2 D^3} \left[(D^2 + 16\mu^2 D_h^2)^{3/2} - D^3 \right]$$

This calculated stress shall not exceed 2/3 of the yield strength of the bolt material. Refer to 3.3.6.2 for additional requirements on threaded fasteners securing waterbox inspection openings on class 1 coolers.

Note: P in equation above is preload force using the low coefficient of friction (0.065) and the installation torque calculated above.

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(5) Fatigue analysis of bolts shall be in accordance with section III (class 1 vessel) of ASME Boiler and Pressure Vessel Code.

(d) For class 1 coolers, all submergence pressure boundary studs and bolts shall be of a reduced shank construction to reduce bending stresses.

3.1.10 Shockproofness. Air coolers shall withstand the high-impact shock test as specified for grade A, class I equipment of MIL-S-901.

3.1.10.1 Coolers supported by the parent machine. Coolers that are to be supported by the parent machine shall withstand high-impact shock tests (see 4.2.3).

3.1.10.2 Coolers constructed to be separately mounted. Coolers constructed for separate mounting shall pass high-impact shock tests (see 4.2.3).

3.1.10.3 Shock-resistant construction. Clearance for fasteners shall be no greater than the following, unless permitted by the applicable pipe flange standard:

Nominal bolt diameter (inches)	Nominal diameter of hole (inches)
3/4 and smaller	Nominal bolt diameter plus 1/32
Larger than 3/4	Nominal bolt diameter plus 1/16

3.1.11 Water velocity and pressure drop limits. In the construction of cooling water spaces and connections, a smooth flow path shall be provided in order to minimize erosion-corrosion attack. Cooling water velocity at design point shall not exceed 7.5 feet per second at the flange face of the entrance nozzle or 6.0 feet per second through the tubes. For coolers using sea water as a coolant, the velocity through the tubes shall be not less than 4.5 feet per second. Unless otherwise specified (see 6.2.1), the maximum permissible pressure drop across the tube side shall be 6 pounds per square inch (lb/in^2).

3.1.12 Water inlet temperature. Unless otherwise specified for design purposes (see 6.2.1), temperature of coolant supply shall be taken as follows:

		Coolant	Temperature degrees Fahrenheit (°F)
(a)	Submarines:	Sea water	85
		Fresh water	95
(b)	Surface ships:	Sea water	90
		Fresh water	95

3.1.13 Waterside pressures. The waterside design pressure shall be as specified (see 6.2.1) and shall be a minimum of 50 lb/in^2. The hydrostatic test pressure shall be 1-1/2 times the design pressure.

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3.1.14 Heat transfer surface. The amount of surface provided shall be based on the duty specified (see 6.2.1), applying a 10 percent fouling factor to the heat transfer coefficient for clean tubes. When the machine is operating at rated electrical load, the air cooler shall cool the air so that the temperature of the air leaving the cooler shall not exceed 122°F, unless otherwise specified (see 6.2.1).

3.1.15 Inlet and outlet water connections. Inlet and outlet water connections shall be flanged, and the flanges shall be cast integrally or of the welding type. Unless otherwise specified (see 6.2.1), requirements for flanged connections on waterboxes shall be as follows:

(a) Flanges shall be in accordance with the applicable schedule of MIL-STD-777 and MIL-STD-438 for surface ship and submarine air coolers, respectively, in regard to diameter, minimum thickness, drilling, and flange finish.
(b) When flanges using O-ring gaskets are furnished, the groove for the O-ring shall be in the companion flange on the shipbuilder's piping.

3.1.16 Waterside vents and drains. Provision shall be made to permit complete venting and draining of the cooling water circuits. Waterside vent and drain connections shall be a minimum of 3/8-inch nominal pipe size (nps) and shall be at least one nps larger than the tell-tale system vent and drain connections (see 3.3.1.1). Such root connections shall have the wall thickness in accordance with MIL-T-16420, 70-30 copper nickel tube (class 6000 for 1 inch and smaller; class 3300 for 1.050 inch and larger) and shall be of the following types:

(a) For waterside of submarine cooler applications constructed for salt water cooling, a full penetration weld with joint construction in conformance to P-70, P-71, or P-72 of MIL-STD-22 shall be used. Nipples shall be composition 70-30 copper-nickel, in accordance with MIL-C-15726.
(b) For other applications, in addition to the method specified in (a) above, the following are acceptable:

    (1) Single-J, fillet-reinforced weld V-27 in accordance with MIL-STD-22. Nipples shall be copper-nickel conforming to MIL-T-16420.
    (2) Tapped hole in accordance with MS16142 fitted with an adapter of composition 70-30 or 90-10 copper-nickel alloy in accordance wth MIL-C-15726 (see 3.3.1.1).

(c) In addition to methods of (b) above, for other applications where the connection will not normally be in contact with salt water, a socket weld using a T by 2T minimum fillet in accordance with P-14 of MIL-STD-22 will be acceptable.

3.1.16.1 High pressure applications. Preferred practice for high pressure applications is to lead waterbox vent and drain connections out through a waterbox flange or tube sheet.

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3.1.17 Mounting. Cooler shall be mounted such that any leakage from the joint between the outer tube sheets and the waterboxes will not enter the attached equipment. This joint shall be visible from the exterior of the machine without removal of any covers or other machine parts.

3.1.18 Supports. The means of support for air coolers shall be independent of connecting piping. If coolers are supported from a vertical structure, the supports shall be constructed so that bolts in shear will not constitute the primary means of support. Coolers shall not be supported by plates or brackets in such a manner that the primary means of support is obtained from the bolts securing the frame-tube sheet-waterbox flange joints.

3.1.19 Gasket application. For waterside application of 200 lb/in^2 design pressure and under, flat gaskets may be used. O-ring gaskets shall be used for design pressures above 200 lb/in^2. Use of blind gaskets or packings (those not capable of being replaced without destructive disassembly) is prohibited.

3.1.20 Flat gaskets. Flat gaskets shall be 1/16-inch thick. Coolers shall be constructed so that an adequate gasket width is provided under waterbox partitions, and this gasket width shall be not less than 5/16 inch. Gasket material shall be in accordance with MIL-R-6855, class 1, 80 durometer. Synthetic rubber shall be used, except for applications where design pressure does not exceed 100 lb/in^2, MIL-G-1149, type II, class 5, synthetic rubber may be used.

3.1.21 O-ring gaskets. O-ring gaskets shall conform to type I of MIL-G-21610, except for the special applications for adaptors and plugs. In these cases, O-ring gaskets in accordance with class 2 of MIL-R-83248 shall be used. O-ring segments for sealing under waterbox partitions shall be integral with the peripheral O-ring. Design of retaining grooves for O-ring gaskets shall conform to MIL-G-5514.

3.1.22 Identification plates. Each cooler shall bear an identification plate in conformance to type A, B, C or D of MIL-P-15024 and MIL-P-15024/5 with material choice being limited to the wrought brass, cast brass, cast bronze or corrosion-resisting steel prescribed therein. (NOTE: All plates shall be normal service plate.) Provision shall be made for the following information:

(a) Name of unit (GENERATOR AIR COOLER, MOTOR AIR COOLER).
(b) Class of unit cooler (see 1.2).
(c) Manufacturer's service part number.
(d) National stock number. (Allow 20 spaces.)
(e) Name of manufacturer.
(f) Contract or order number. (The contracting activity's contract or purchase order number shall be entered in this space.) (Utilize width of plate to allow maximum number of spaces.)
(g) Blank space for Defense Contract Administration Services Management Area (DCASMA) stamp.
(h) Date of manufacture.
(i) Serial number.
(j) Maximum test pressure, leak-off system.

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(k) Maximum hydrostatic test pressure, waterside. (For class 1 coolers, no entry shall be made in this space (see 3.1.13).)
(l) Four blank spaces for unit number. (This space will be used for numbering for shipboard reference purposes when required, the stamping to be done by shipyard.)
(m) Designation "U.S.".

Design pressure entries shall not appear on identification plates.

3.2 Materials. The materials as specified in 3.2.1 through 3.2.4 shall be used in the construction of the coolers, except that the manufacturer shall have the option of substituting commercial material having equal or better physical and chemical properties than the materials specified under Federal or Military specifications.

3.2.1 Class 1 cooler materials. Class 1 coolers shall be constructed of the materials specified in table I.

TABLE I. Class 1 cooler materials.

Part	Material	Applicable document
(a) Airside materials (frame, support plates, cover plates, and so forth)	Steel, grade A, class U Steel, grade B, seamless	MIL-S-22698 ASTM A 106
(b) Waterboxes[1]	Copper-nickel-alloy, composition 70-30 Tubing, copper-nickel-alloy, composition 70-30 Copper-nickel-alloy, composition 70-30; or bronze, nickel-aluminum	MIL-C-15726 MIL-T-16420 MIL-C-20159 MIL-B-24480
(c) Outer tube sheets	Copper-nickel-alloy, composition 70-30	MIL-C-15726
(d) Inner tube sheets	Copper-nickel-alloy	MIL-C-15726

See footnotes at end of table.

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TABLE I. Class 1 cooler materials. - Continued

Part	Material	Applicable document
(e) Inner tubes	Copper-nickel-alloy, composition 70-30	MIL-T-15005
(f) Outer tubes	Copper	MIL-T-24107
(g) Fins	Nonferrous	----
(h) Bolts or studs in joints involving sea water tightness or subject to submergence pressure[4] [5]	Nickel-copper-aluminum alloy, hot finished, annealed, and age hardened, or cold drawn, annealed, and age hardened	QQ-N-286
(i) Bolts or studs for other service	Nickel-copper, grade 400; or nickel-copper-aluminum, grade 500	MIL-S-1222
(j) Nuts for bolts and studs specified in (h)[5]	Nickel-copper-aluminum alloy Nickel-copper alloy, class A, hot finished Nickel-copper alloy, cold drawn and stress relieved, or hot finished	QQ-N-286 QQ-N-281 MIL-N-24106
(k) Nuts: For use on air side	Steel, nylon insert; or nickel-copper alloy, nylon insert	MS17829 MS17828
(l) Nuts for bolts and studs specified in (i)[5]	Nickel-copper, grades 400 and 405	MIL-S-1222
(m) Jack screws[5]	Copper-aluminum alloy, copper alloy no. 614; or phosphor bronze; or copper-silicon alloy	QQ-C-465 QQ-B-750 QQ-C-591
(n) Zinc protectors	Zinc	MIL-A-18001

See footnotes at end of table.

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TABLE I. Class 1 cooler materials. - Continued

Part	Material	Applicable document
(o) Vent and drain nipples	Copper-nickel alloy, seamless, composition 70-30	MIL-T-16420
(p) O-ring gaskets[2]	Rubber, synthetic, type I Rubber, synthetic, class 2	MIL-G-21610 MIL-R-83248
(q) Tell-tale system vent and drain connectors[3]	Copper-nickel alloy composition 70-30	MIL-C-15726

[1] Inspection covers including zinc covers shall be of the same copper alloy as the waterboxes to which fitted.
[2] If required, lubricant for O-ring gaskets shall be in accordance with DOD-G-24508 or MIL-S-8660.
[3] See 3.3.1.1 and figure 1.
[4] Bolts and studs shall also conform to MIL-S-1222.
[5] Form in accordance with MIL-S-1222.

3.2.2 Class 2 cooler materials. Class 2 coolers shall be constructed of any of the materials specified in table I or those specified in table II except that airside materials shall be as specified in table I.

TABLE II. Additional class 2 cooler materials.

Part	Material	Applicable document
(a) Plugs, zinc support[2]	Copper-nickel alloy, composition 70-30	MIL-C-15726
(b) Gaskets[1]	Rubber, synthetic, class 1, grade 80 Rubber, synthetic type II, class 5	MIL-R-6855 MIL-G-1149

[1] See 3.1.20.
[2] Plugs shall conform to Drawing 803-5959186 (see 3.3.5.4).

3.2.3 Class 3 cooler materials. Class 3 coolers shall be constructed of the materials specified in table III except that airside materials shall be as specified in table I.

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TABLE III. Class 3 cooler materials.

Part	Material	Applicable documents
(a) Waterboxes[1]	Copper-nickel alloy Tubing, copper-nickel alloy Copper-nickel alloy Bronze, nickel-aluminum; or Bronze, sand castings, alloy C90300, C90500 or C92200	MIL-C-15726 MIL-T-16420 MIL-C-20159 MIL-B-24480 QQ-C-390
(b) Outer and inner tube sheets	Copper-nickel alloy	MIL-C-15726
(c) Inner tubes	Copper-nickel alloy, composition 90-10	MIL-T-15005
(d) Outer tubes	Copper	MIL-T-24107
(e) Fins	Nonferrous	-----
(f) Bolts, studs, and nuts	Nickel-copper, grades 400 and 405, or nickel-copper-aluminum, grade 500	MIL-S-1222
(g) Nuts for use on airside	Steel, nylon insert; or Nickel-copper alloy, nylon insert	MS17829 MS17828
(h) Jack screws	Copper-aluminum alloy, copper alloy no. 614; phosphor bronze; or copper-silicon alloy	QQ-C-465 QQ-B-750 QQ-C-591
(i) Zinc protectors	Zinc	MIL-A-18001
(j) Gaskets[2]	Rubber, synthetic, class 1, grade 80 Rubber, synthetic, type II, class 5	MIL-R-6855 MIL-G-1149
(k) Plugs, anode support	Copper-nickel alloy, composition 70-30 or 90-10	MIL-C-15726

See footnotes at end of table.

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TABLE III. Class 3 cooler materials. - Continued

Part	Material	Applicable document
(l) Washers	Copper-aluminum alloy, copper alloy no. 614; or brass, naval	QQ-C-465 QQ-B-639
(m) Adaptors	Copper-nickel alloy	MIL-C-15726
(n) O-ring gaskets[4]	Rubber, synthetic, type I Rubber, synthetic, class 2	MIL-G-21610 MIL-R-83248
(o) Tell-tale system vent and drain connectors[3]	Copper-nickel alloy	MIL-C-15726

[1] Inspection covers including zinc covers shall be of the same copper alloy as the waterboxes to which fitted.
[2] See 3.1.20.
[3] See 3.3.1.1 and figure 1.
[4] If required, lubricant for O-ring gaskets shall be in accordance with DOD-G-24508 or MIL-S-8660.

3.2.4 Class 4 cooler materials. Class 4 coolers shall be constructed of the materials specified in table IV except that airside materials shall be as specified in table I.

TABLE IV. Class 4 cooler materials.

Part	Material	Applicable document
(a) Waterboxes	Copper-nickel alloy Tubing, copper-nickel alloy copper-nickel alloy Bronze, nickel-aluminum; or bronze, sand castings, alloy C90300, C90500 or C92200	MIL-C-15726 MIL-T-16420 MIL-C-20159 MIL-B-24480 QQ-C-390
(b) Outer and inner tube sheets	Copper-nickel alloy, or nickel-aluminum-bronze, alloy C63000, or aluminum-bronze, alloy C61400	MIL-C-15726 ASTM B 171
(c) Inner tubes	Copper-nickel alloy, composition 90-10	MIL-T-15005
(d) Outer tubes	Copper	MIL-T-24107
(e) Fins	Nonferrous	-----

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TABLE IV. Class 4 cooler materials. - Continued

Part	Material	Applicable document
(f) Nuts, studs, and nuts	Nickel-copper, grades 400 and 405, or nickel-copper-aluminum, grade 500	MIL-S-1222
(g) Nuts for use on air side	Steel, nylon insert; or Nickel-copper alloy, nylon insert	MS17829 MS17828
(h) Jack screws	Copper-aluminum alloy, copper alloy no. 614; or Phosphor bronze; or Copper-silicon alloy	QQ-C-465 QQ-B-750 QQ-C-591
(i) Gaskets[1]	Rubber, synthetic, class 1, grade 80 Rubber, synthetic, type II, class 5	MIL-R-6855 MIL-G-1149
(j) Washer	Copper-aluminum alloy, copper alloy no. 614; or brass, naval	QQ-C-465 QQ-B-639
(k) Adaptors	Copper-nickel alloy	MIL-C-15726
(l) O-ring gaskets[2]	Rubber, synthetic, type I Rubber, synthetic, class 2	MIL-G-21610 MIL-R-83248
(m) Tell-tale system vent and drain connections[3]	Copper-nickel alloy	MIL-C-15726

[1] See 3.1.20.
[2] See footnote on O-ring lubricant in table I and III.
[3] See 3.3.1.1 and figure 1.

3.2.5 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

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3.3 Construction.

3.3.1 Double tube and tell-tale provision. Air coolers shall have extended surface outer tubes sealed into inner tube sheets or headers by brazing, welding or rolling. These outer tubes shall have longitudinal grooves at least 0.018-inch deep in their internal surfaces. No more than 50 percent of the outer tube inner-diameter shall be in contact with the inner tube outer-diameter after the inner tubes are expanded tightly into the outer tubes. After inner tube expansion, there shall be adequate contact between the two tubes to conduct the heat, and the grooves will readily lead off leakage from any perforation of an inner tube to the end compartments formed between outer tube sheets and inner tube sheets (headers). Such leakage shall be readily detected through the use of a tell-tale system leading from these end compartments with vent and drain connections on each outer tube sheet. These connections will terminate in straight threaded O-ring connections conforming to MS16142 (see 3.3.1.1). These connections shall be located outside the motor or generator so that any leakage past these connections will not get into shell or air side of the cooler. These connections shall not be plugged except while leak testing the void space. The void space between inner and outer tubes and between inner tube sheet (or header) and outer tube sheet shall be sealed by the tube-to-tube sheet joints as required above and by welding the inner and outer tube sheets. Any hole or other opening which penetrates either tube sheet, other than holes for heat transfer tubes and vents and drains, shall be located outside the weld between tube sheets. Fasteners shall not be welded or brazed to tube sheets. The tell-tale system of the cooler shall withstand an air pressure test of not less than 10 lb/in^2 with the waterbox removed so that the location of leaking inner tubes or inner tube/outer tubesheet joints can be determined. Hardware that is not normally installed shall not be required to pass the air pressure test. An instruction plate conforming to MIL-P-15024 and MIL-P-15024/5 shall be provided to indicate the function of these holes and to warn against improper plugging of them. This plate shall be located close to one of the leak-off connections.

3.3.1.1 Tell-tale system vents and drains. At each end of coolers having an odd number of waterside passes and at the inlet-outlet end of coolers having an even number of waterside passes, the required vent and drain connections for the tell-tale system shall be made by the method specified in 3.1.16 (b)(2), except that minimum size shall be 1/4-inch nps and shall be at least one nps smaller than the waterside vent and drain connections (see 3.1.16). O-ring connectors in accordance with figure 1 shall be furnished for each tell-tale system connection.

3.3.2 Tubes. Heat transfer tubes shall be straight and seamless.

3.3.2.1 Tube sizes. Tube sizes shall be as follows:

(a) Unless otherwise specified (see 6.2.1), class 1 coolers shall be provided with inner tubes of 5/8-inch outside diameter (od), 0.065-inch (number 16 Birmingham wire gauge (BWG)) wall thickness.
(b) Class 2 and 3 coolers shall be provided with inner tubes of 5/8-inch od, 0.049-inch (number 18 BWG) wall thickness or 1/2-inch od, 0.049-inch (number 18 BWG) wall thickness.

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(c) Class 4 coolers shall be provided with inner tubes of 5/8-inch od, 0.049-inch (number 18 BWG) wall thickness; 1/2-inch od, 0.049-inch (number 18 BWG) wall thickness; or 3/8-inch od, 0.035-inch (number 20 BWG) wall thickness.

3.3.2.2 Tube expansion in outer tube sheets. Tubes shall be expanded into the outer tube sheets at both ends by means of a tube expander in accordance with MIL-E-15809 that shall be governed by an automatic tube expander control. The expanded portion of the tube-to-tube sheet joint shall not approach closer than 1/8 inch to the back face of outer tube sheet. The minimum depth of expansion of tubes shall be as specified in table V.

TABLE V. Minimum depth of expansion.

Tube od (inch)	Minimum depth of expansion (inch)
5/8	5/8
1/2	5/8
3/8	1/2

3.3.3 Outer tube sheets.

3.3.3.1 Minimum thickness. Tube sheet minimum thickness shall be not less than:

(a) For joints with inlet end flare (see 3.3.3.3) - Depth of expansion (see 3.3.2.2), plus depth of flare, plus 1/8 inch.
(b) For non-flared joints - Depth of expansion plus 1/8 inch.

3.3.3.2 Tube holes. Holes for tubes in outer tube sheets shall be drilled 0.001 inch larger than the od of the inner tube (after assembly with the outer tube) with a plus tolerance of 0.005 inch.

3.3.3.3 Flaring of holes. The holes for the inlet ends of the tubes shall be flared in accordance with table VI to allow for belling the ends of the tubes.

TABLE VI. Flaring of holes for inlet ends.

Tube od (inch)	Radius of flare (inch)	Diameter of flare, at outside face of tube sheet (inch)
5/8	1/2	3/4
1/2	3/8	5/8
3/8	5/16	1/2

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3.3.3.4 Grooving of holes in tube sheets. The holes for tubes in the tube sheets shall be grooved in accordance with table VII.

TABLE VII. Grooving of holes in tube sheets.

Tube size od	Cooler application	Number of grooves in outer tube sheet[1]	Number of grooves in inner tube sheet[2]	Width of groove (inch)	Depth[3] of groove (inch)	Space between grooves (inch)
5/8 inch and 1/2 inch	Class 1 or 2 cooler	3	1	1/16	0.012	1/8
	Class 3 or 4 cooler	2	1	1/16	0.012	1/8
3/8 inch	Class 4 cooler	1	1	1/16	0.012	---

[1] The grooves shall be located in the expanded portion of the joint.
[2] No grooving of inner sheet is required when a welded or brazed joint is used for outer tube-to-inner tube sheet attachment.
[3] Tolerance on groove depth is plus or minus 0.005 inch.

3.3.3.5 Rounding of tube hole edges. Where tubes are to be expanded the edges of the holes for the tubes shall be rounded on a 1/16-inch radius except where a belled contour is required for the tube inlet end.

3.3.3.6 Inner tubes inlet ends. The inlet ends of inner tubes shall be expanded and flared (see table VI) and the ends shall then be finished flush with the face of the tube sheet. In no case shall the ends of tubes be below the face of the tube sheets. Discharge ends of tubes shall not protrude more than 3/16-inch beyond the face of the tube sheet.

3.3.4 Support plates. Holes for tubes in support plates shall be not more than 1/64-inch greater (nominal diameter) than the od of the expanded tube. Edges of the holes shall have all roughness removed.

3.3.5 Waterboxes. Waterboxes shall be proportioned to provide sufficient area at all points for flow of the circulating water with the minimum of turbulence and for uniform distribution of the water of all tubes in each pass. Components shall have an inlet waterbox compartment depth, measured normal to the tube sheet, of not less than one-third the diameter of a circle of area equivalent to that of the tube sheet exposed to circulating water flow. Interpass (reversing) waterbox compartments shall be deep enough to ensure that the cross section of full flow shall have an area not less than four times the area of the tubes discharging into the compartment. Waterboxes shall be provided with vent holes in the partitions between passes.

3.3.5.1 Waterbox removal. Removal of the waterboxes shall only require removal of the waterbox fasteners. Removal of covers or any other parts of the cooler or parent equipment shall not be required to remove the waterboxes.

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3.3.5.2 Waterbox partition plate. Waterbox partition plate width shall be not less than 1/4 inch at the gasket contact surface.

3.3.5.3 Jack screws. Jack screws shall be fitted on the waterboxes of coolers of 12-inches and above nominal waterbox diameter.

3.3.5.4 Corrosion preventive anodes. Anodes shall be provided for salt water cooled air coolers in accordance with MIL-A-19521 with the following exceptions:

(a) For class 1 air coolers, threaded support plugs shall not be used. Cover plates secured by threaded fasteners shall be used for support of anodes.
(b) For class 2 air coolers, bolted cover plates or threaded plugs shall be used for support of anodes. If used, threaded plugs shall conform to NAVSEA 803-5959186.

3.3.6 Cyclic life design.

3.3.6.1 Class 1 coolers. For class 1 coolers, the waterside shall withstand cyclic loading as specified (see 6.2.1) with associated temperature variation in conjunction with piping reaction load below and 150 cycles of the specified hydrostatic test pressure when applied in the waterbox. Diameter and thickness of attached circulating water piping shall be as specified (see 6.2.1). Piping reaction values shall be as follows:

$$\text{Piping axial load} = \pi r_m t \left(s_y - \frac{Pr_i}{2t} \right)$$

$$\text{Piping bending moment} = 1.3 \pi r_m^2 t \left(s_y - \frac{Pr_i}{2t} \right)$$

$$\text{Piping torsional moment} = \pi r_m^2 t \sqrt{s_y^2 - \left(\frac{Pr_i}{2t} \right)^2}$$

Where:

P = design pressure for the component, lb/in^2

r_i = pipe inside radius, inches

r_m = pipe mean radius, inches $\left(\frac{r_i + r_o}{2} \right)$

r_o = pipe outside radius, inches

s_y = tabulated value of yield strength of the piping material at the component's design temperature, lb/in^2

t = pipe thickness, inches

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3.3.6.2 Threaded fasteners. Threaded fasteners securing waterbox inspection openings on class 1 coolers shall be constructed so that 200 inspection cover removal and reinstallation cycles and cyclic loading specified in 3.3.6.1 will not cause the maximum cumulative usage factor to exceed 0.8 (see 3.1.9.1.6(c)(5)).

3.3.6.3 Threaded plugs, connections, and closures. Threaded plugs, threaded connections and threaded closures are prohibited in the submergence pressure boundary of class 1 coolers.

3.3.7 Cyclic life analysis. Except where other analytical methods are permitted herein, boundaries subject to cyclic loading of class 1 coolers shall be analyzed for specified load combinations (for example, internal pressure in combination with that piping reaction load that produces the highest stress) in accordance with the methods and criteria of the latest edition (including revisions, addenda, and applicable cases) of section III of the ASME Boiler and Pressure Vessel Code, class 1 vessels. Design fatigue curves for materials commonly used in fabrication of coolers and design stress intensity values for materials not covered in section III (class 1 vessels) of ASME Boiler and Pressure Vessel Code shall be obtained from NAVSEA. Method of analysis for flat, perforated plates shall be in accordance with section III of ASME Boiler and Pressure Vessel Code. Cumulative usage factor for all transients shall not exceed 0.8.

3.3.7.1 Design report. When specified in the contract or order, a design report shall be prepared (see 6.2.2 and appendix A).

3.4 Painting. Coolers shall be painted as follows:

(a) External and internal nonferrous surfaces shall not be painted.
(b) Steel surfaces shall be treated as follows:

(1) After all machining, welding, and brazing operations are completed, all rust and other corrosion products and flux shall be removed by abrasive blasting, sanding, wire brushing, or other mechanical means. This surface shall be thoroughly cleaned of all grease, oil, and dirt by solvent wiping and rinsing, vapor degreasing, or caustic washing followed by rinsing.
(2) Primer shall be applied in accordance with one of the following methods:

a. One coat of primer pretreatment in accordance with DOD-P-15328 (0.0002 to 0.0005-inch dry film thickness) followed by primer conforming to TT-P-1757 or TT-P-664.
b. A hot-dip tank phosphate treatment conforming to type I of TT-C-490 (0.0006 to 0.001-inch dry film thickness) followed by primer conforming to TT-P-664 or TT-P-1757.
c. Primer conforming to TT-P-1757 or TT-P-664 shall be applied as one coat (0.0006 to 0.0008-inch dry film thickness).

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(3) Two coats of gray enamel conforming to class 2 of MIL-E-15090 shall be applied as continuous films approximately 0.001-inch thick each.

3.5 Aperture tabulating cards. When specified in the contract or order, aperture tabulating cards shall be prepared (see 6.2.2).

3.6 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2 and appendix B).

3.7 Certification data/report. When specified in the contract or order, a certification data/report shall be prepared (see 6.2.2 and appendix C).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Critical dimensions. For class 1 coolers, critical dimensions such as waterbox thicknesses, tube sheet thicknesses, tube hole sizes, and tube sheet ligaments shall be measured. A critical dimension is defined as a dimension based on fatigue analysis or directly related to another dimension so based (for example, the size of a tube hole). Departures from limits established by accepted drawings shall be classed as major variations. Approval consideration of major variations shall be only by the drawing review activity (see 6.5), and a prerequisite for approval consideration will be supporting calculations demonstrating the effect of the error on the fatigue life. For class 1 coolers, a 100 percent examination of the tube sheet holes and ligament widths shall be made.

4.1.3 Critical materials. For class 1 coolers, the following items are designated as critical materials. Each of these items shall have material traceability including heat lots, chemical and physical test reports. Critical materials shall consist of the following:

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(1) Outer tube sheets.
(2) Sea water vent and drain connections.
(3) Heat transfer tubes (inner tubes only).
(4) Waterboxes including all parts, subparts, and access covers.
(5) Nuts, studs, bolts and washers, joining sea water containing parts.

Traceability records shall be retained by the contractor for 7 years from delivery. At the expiration of the retention period, all records shall be made available to NAVSEA or its authorized representative by written notification. If no disposition is authorized within 6 months, the records may be destroyed.

4.2 Tests. Except for the high impact shock tests, those tests that require assembly with the parent machine for which the cooler is constructed shall be conducted at the plant of the parent machine manufacturer. It shall be the responsibility of the prime contractor to ensure that tests required on assembled units are made. The manufacturer shall make sufficient tests to ensure that the construction of the cooler conforms to this specification.

4.2.1 Tests at the place of manufacture. Each air cooler shall be given the following pressure tests:

	Test pressure
Waterside (hydrostatic):	See 3.1.13
Leak-off system (with air):	10 lb/in^2 (minimum)

Pressure tests shall be held for 30 minutes or for sufficient time to examine all external surfaces, whichever is longer. Acceptance criteria for tests shall be no leakage.

4.2.2 Tests at plant of parent machine manufacturer. Where tests are required by the specifications for the parent machine on a selective or periodic basis, these shall be conducted with the air cooler installed. When performance tests of the parent machine are required, the cooler shall be supplied with cooling water at rated flow and temperature. Measurements shall include the following:

(a) Water flow rate (gal/min).
(b) Water inlet temperature (°F).
(c) Water outlet temperature (°F).
(d) Water pressure drop across cooler (lb/in^2).
(e) Air inlet temperature (°F).
(f) Air outlet temperature (°F).
(g) Air pressure drop across cooler (inches H_2O).

4.2.2.1 Calculated values. From the data of 4.2.2, the following values shall be calculated for comparison with values shown on the ship's drawing:

(a) Logarithmic mean temperature difference (°F).
(b) Heat transfer capacity (Btu per hour).
(c) Heat transfer rate (clean tube condition) (Btu per hour per square foot outside surface per °F logarithmic mean temperature difference).

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4.2.3 High impact shock test. The test shall be as specified for grade A, class I in MIL-S-901 and as specified herein:

(a) When the parent equipment is being shock tested, the air cooler shall be shock tested in its installed position.

(b) The waterside of the air cooler shall be filled with fresh water and pressurized to design operating pressure during the test. All equipment or parts shall be mounted on the shock machine in a manner simulating the actual installation on board ship. The weight designation of the shock test shall be as required by the combined weight of the equipment being tested. When the cooler is being tested independent of the parent equipment, grade 2 bolts in accordance with MIL-S-1222 shall be used to hold down the cooler during the shock test.

(c) Failure to perform principal function. Any of the following failures shall be considered as cause for rejection of the equipment:

(1) Breaking of any parts.
(2) Appreciable distortion or dislocation of any parts such as mounting feet, tubes, baffles or attached piping.
(3) Leakage of water from the cooling waterside of the cooler.
(4) Following the shock test, the leak-off side of the cooler shall be pressurized with air to design test pressure. There shall be no pressure drop indicating leakage in the leak-off portion of the cooler.

(d) Disposal of shock-tested equipment.

(1) Equipment which has been subjected to the high-impact shock test and has failed to perform any of the principal functions specified herein shall not be acceptable, either in whole or any of the parts.
(2) Equipment which has been subjected to the high-impact shock test and has successfully performed all the principal functions specified herein will not be considered for acceptance until it is first disassembled, inspected and refurbished.

(e) The place at which the shock test shall be conducted. Shock tests may be performed at the contractor's plant, commercial laboratory, or Government laboratory which is suitably equipped to perform these tests. In either case, the tests shall be conducted at the expense of the contractor. Where shock tests are conducted at a Government laboratory, copies of all master drawings shall accompany the equipment.

(f) The number of air coolers to be tested. Unless otherwise specified in the contract or order (see 6.2.1), an air cooler of each size and similar construction shall be shock tested.

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4.3 Proof test - special case (cyclic loading stipulated). For the special case of an inlet-outlet waterbox not amenable to analysis for determination of adequacy for cyclic life requirements, the adequacy of the waterbox or a scale model thereof shall be verified by a proof test. This proof test shall consist of a qualitative survey of the waterbox utilizing brittle coating (stress coat or its equivalent), or photostress (molded birefringent coatings) techniques as deemed suitable, in order to determine the location of maximum strain. Quantitative measurement of the magnitude of maximum principal strain in the waterbox shall be obtained from strain gauges of 1/8-inch gauge length or smaller. Waterbox loadings shall be comprised of a combination of design submergence pressure in conjunction with piping reaction load (that load which will produce highest stress) applied to the nozzles. The nozzle loads shall be calculated as specified in 3.3.6.1. A loading jig utilizing mechanical or hydraulic jacks shall be provided to apply external loads to the nozzle flanges. The following combinations of loads shall be applied:

(a) Internal pressure and external piping load (that piping reaction load which produces the highest stress) applied to one flange.
(b) Internal pressure and equal external loads (that piping reaction load which provided the highest stress) applied to both flanges.

Since the angular orientations of the applied bending moments to give the maximum stress for each load combination are unknown, and the location of maximum stress is also unknown, the angular orientations of the applied bending moments for load combinations (a) and (b) shall correspond to increments of 45 degrees around the nozzles from 0 to 360 degrees in an initial run, and a minimum of 9-degree increments shall be used within the 45-degree maximum stress sector in a follow-on run. The location, magnitude and direction of the maximum principal strain (stress) can be determined by brittle coating or photostress, but must be checked with electric resistance strain gauges. If the maximum range of strain (stress) intensities thus determined in any location of the waterbox (by either load combination (a) or (b)) does not exceed the allowable strain (stress) intensity range for the stipulated cycle life as determined by Fatigue Curve for the particular material involved, the waterbox design is acceptable. When specified in the contract or order, test reports shall be prepared (see 6.2.2). Approval of the procedures and proposed set-up is required prior to testing. The DCASMA will generally monitor the set-up for test, adherence to procedures, and the taking of data. Certification that a proof test in accordance with the above has been previously performed on equipment identical to that being furnished, may be submitted as a basis for requesting a waiver.

4.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2.1).

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5.1.1 Level A. Coolers shall be preserved in accordance with method III of MIL-P-116. Openings shall be sealed with caps, plugs or barrier material conforming to MIL-B-121 secured in place with tape.

5.1.2 Commercial. Coolers shall be preserved in accordance with ASTM D 3951, including the requirement that all openings shall be sealed to prevent entrance of foreign matter.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2.1).

5.2.1 Level A. Coolers preserved in accordance with 5.1 shall be packed in wood-cleated plywood or nailed wood boxes or sheathed crates conforming to PPP-B-601 (overseas type), PPP-B-621 (class 2), MIL-C-104, as applicable. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds, unless the weight of a single cooler exceeds this weight. Wood-cleated plywood and nailed wood boxes shall be used for gross weights not exceeding 500 pounds. The boxes shall be modified by the addition of skids and the coolers bolted through the skids. When the gross weight of the unit exceeds 500 pounds, the cooler shall be packed in a sheathed crate.

5.2.1.1 Waterproofing. Equipment bolted to the base of a wood or wood-cleated box shall be shrouded within the box with barrier material conforming to PPP-B-1055. Shrouding shall be applied at least two-thirds of the way down in a manner to prevent the formation of water pockets and to permit free circulation of air. Sharp projections which would tend to tear the shroud shall be padded or covered with cushioning material.

5.2.2 Level B. Coolers preserved in accordance with 5.1 shall be packed in wood-cleated fiberboard, wood-cleated plywood, nailed wood boxes or crates conforming to PPP-B-591, PPP-B-601 (domestic type), PPP-B-621 (class 1), MIL-C-3774, and MIL-C-52950. The gross weight of wood and wood-cleated boxes shall not exceed 200 pounds unless the weight of a single item exceeds this weight. Boxes shall be modified for gross weights over 200 pounds by the addition of skids, and the cooler shall be bolted through the base and skids. Coolers packed in unsheathed crates shall be shrouded as specified in MIL-C-52950 and appendix thereto.

5.2.3 Commercial. Coolers preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2.1), level A and B interior packages and shipping containers shall be marked in accordance with MIL-STD-129, and commercial packages and shipping containers shall be marked in accordance with ASTM D 3951. Equipment serial number shall be marked on one face of the shipping container.

6. NOTES

6.1 Intended use. The air coolers covered by this specification are intended for heat transfer equipment used to cool the air circulating within electrical equipment such as motors and generators installed on Naval ships.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class of cooler (see 1.2).
(c) For submarine air coolers, dimensions of hatch, bulkhead opening, or other access openings available for removal of cooler from the compartment in which the unit will be installed (see 3.1.2.1).
(d) Ship motion and attitude requirements, if other than specified (see 3.1.3).
(e) Rate of flow of air to be cooled, cubic feet per minute (ft^3/min).
(f) Maximum permissible pressure drop across the waterside (lb/in^2), if other than specified (see 3.1.11).
(g) Inlet temperature of water (°F), if other than specified (see 3.1.12).
(h) The waterside design working pressure, lb/in^2 (see 3.1.13 and 4.2.1).
(i) Heat transfer duty, Btu/hr (see 3.1.14).
(j) Air outlet temperature (°F), if other than specified (see 3.1.14).
(k) Rate of water flow, if established (gal/min).
(l) Requirements for inlet and outlet water connections, if other than specified (see 3.1.15).
(m) For class 1 coolers, the diameter and thickness of inner tubes, if other than specified (see 3.3.2.1).
(n) For class 1 coolers, the diameter and thickness of attached circulating water piping (see 3.3.6.1).
(o) For class 1 coolers, cyclic life requirements (see 3.3.6.1).
(p) Number of air coolers shock tested (see 4.2.3(f)).
(q) Levels of preservation and packing required (see 5.1 and 5.2).
(r) Special marking requirements (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.3.7.1 and appendix A	Report, design	DI-R-24039	----
3.5	Imaged aperture/tabulating cards	DI-E-20477	----
3.6 and appendix B	Drawings, engineering and associated lists	DI-E-7031	Level 3
3.7 and appendix C	Certification data/report	UDI-A-23264	----
4.3	Reports, test	DI-T-2072	----

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.2.2 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, Military specifications and standards which have been cleared and listed in DoD 5010.12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract. Technical content shall include the requirements of appendix D, titled "Manual Technical Content Requirements."

6.3 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.3.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Drawing review activity. As used herein "drawing review activity" is generally a Government Command or activity such as Naval Sea Systems Command, a Supervisor of Shipbuilding, or an authorized representative. Communication with the drawing review activity should be handled through the contracting activity.

6.6 Cross-reference. The classification of air coolers given in this specification has been revised from previous revision of this specification. Equivalent requirements between this specification and MIL-C-19836A(SHIPS) are as follows:

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MIL-C-19836B	MIL-C-19836A
Class 1	Class 1
Class 2	----
Class 3	Class 2
Class 3	Class 3
Class 4	----

Class 2 and 3 of MIL-C-19836A(SHIPS) are both replaced by class 3 of MIL-C-19836B(SH).

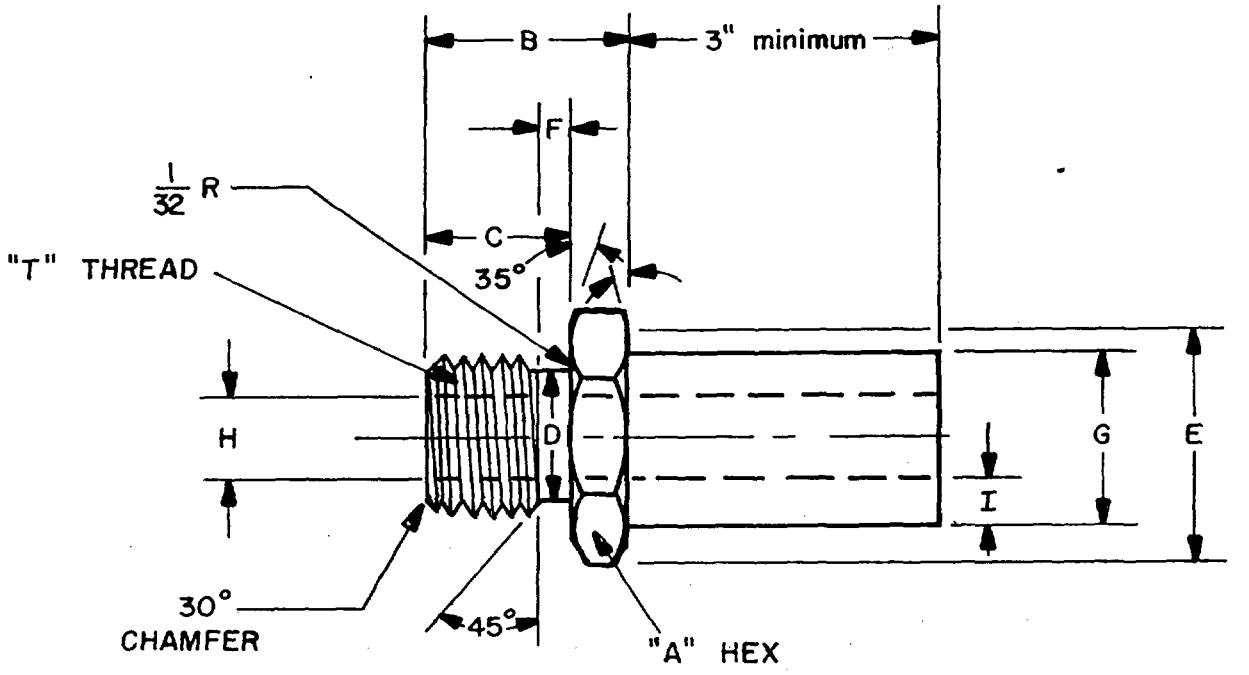
6.7 Subject term (key word) listing.

Double tube
Electrical equipment
Enclosure
Heat transfer
Tell-tale
Waterbox

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4420-N054)

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SH 13043

Dimensions of O-ring connector (all dimensions in inches).

Nominal size	"T" straight thread (UN-2A)	A hexagon across flats	B ± 1/64	C ± 0.005	D ± 0.002 .003	E ± 1/64	F ± 0.005	G ± 0.005	H	I minimum	O-ring AS 568
1/4	7/16-20	9/16	39/64	.360	.364	5/8	.093	.540	.203 ± .003	.120	-904
3/8	9/16-18	11/16	11/16	.391	.482	3/4	.097	.675	.281 ± .004	.148	-906
1/2	3/4-16	7/8	3/4	.438	.660	15/16	.100	.840	.422 ± .004	.203	-908
3/4	1-1/16-12	1-1/4	1	.594	.945	1-5/16	.130	1.050	.656 ± .005	.148	-912
1	1-5/16-12	1-1/2	1	.594	1.195	1-9/16	.130	1.315	.875 $^{+.007}_{-.005}$	.180	-916

Note: The plug portion of this O-ring connector conforms dimensionally to MS18229.

FIGURE 1. O-ring connector for vents and drains.

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APPENDIX A

DESIGN DATA AND CALCULATIONS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the format and content requirements that shall be included in a design report when required by the contract or order. This appendix is mandatory only when the appropriate data item description is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DESIGN REPORT

30.1 Design report content. When required by the contract or order, design report content for applications for which cyclic life requirements are established, shall contain an analysis demonstrating the adequacy of the design. For the class 1 coolers, a design report shall be included. The report shall be in a format to facilitate an independent review of its content. It is, therefore, imperative that it be simple to follow, free from ambiguity, and should contain the following as a minimum:

(a) Description of design requirements such as steady state transient pressures, temperatures, and external loading.
(b) The regions of the component which were analyzed.
(c) The materials which were used and their mechanical properties.
(d) A general description of the methods of analysis and assumptions.
(e) All reference sources shall be listed.
(f) All computer programs shall be properly identified and described.
(g) The report shall contain copies of the computer printouts (input and output).
(h) The stresses shall be tabulated for each area of investigation and compared to the allowable stresses for all stress categories.
(i) Satisfaction of cyclic requirements including stress concentration factors used in the cyclic analysis.

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APPENDIX B

DRAWING TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that shall be included on drawings when required by the contract or order. This appendix is mandatory only when the appropriate data item description is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawings. When required by the contract or order, drawings shall be as specified and contain the following:

(a) Outline drawing: an external arrangement drawing, called "Outline Drawing", shall show all necessary external views of the unit and shall include all external dimensions required for installation thereof, and for connection of the unit to the external piping. The drawing shall show the space required for removal of waterboxes, covers or access openings, anodes (if fitted), location of lifting lugs or eyebolts provided for handling of the unit or its components and size of openings therein provided for lifting, and weights for dry and operating conditions.

(b) Assembly drawing: a drawing showing complete longitudinal and transverse cross-sectional views of the unit shall be called "Assembly Drawing". This drawing shall show the relationship of all parts, direction of flows, arrangement of tubes, and method of tube end expansion and baffles. Liberal use of enlarged views or sections shall be made. If necessary, subassembly drawings conforming to the above may be furnished for individual components of the unit. The drawing or drawings shall be complete so that a thorough understanding of the design and construction of the apparatus may be obtained without reference to related detail drawings.

(1) The assembly drawing shall contain a list of materials showing names of parts with identifying numbers and materials of all parts.
(2) The assembly and outline drawing requirements may be combined on one drawing.

(c) Detail drawings: detail drawings of all major parts such as tube bundle, waterboxes, tube sheets and frame shall be furnished. Drawings shall be completely dimensioned with

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finishes and welding symbols indicated as required for manufacture. In addition, drawings shall contain instructions covering the torqing of bolts for all waterbox and tube sheet bolting. Critical materials shall be designated on the drawings. For class 1 coolers, all submergence pressure boundary weld joints shall be identified by a unique weld joint identification number.

(d) If the manufacturer substitutes commercial material (see 3.2), two columns shall be included in the "List of Materials" on these drawings; one column shall be used for the specified Federal or Military specifications, and the other shall show the commercial specification number for the substituted material. Notes shall then be added to the drawings stating that:

(1) The design is based on the Federal or Military specification material.

(2) The materials listed in the commercial specification column have physical and chemical properties which are equal to or better than the specified Federal or Military specification and may be substituted by the manufacturer.

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APPENDIX C

CERTIFICATION TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the certification technical content requirements that shall be included in the certification data sheet when required by the contract or order. This appendix is mandatory only when the appropriate data item description is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. CERTIFICATION

30.1 Certification data sheet. When required by the contract or order, a certification data sheet shall be as specified and include a tabulation of the following data:

(a) Notes attesting to conformance to the specification, indicating the contract or order, the application (service) of the unit, the number of coolers per ship and per unit of parent equipment.
(b) A "performance data" tabulation to include the following:

(1) Flow rate of air (cubic feet per minute, (ft^3/min)) (specify conditions) and water (gallons per minute (gal/min)).
(2) Inlet temperature of air and water (°F).
(3) Outlet temperature of air and water (°F).
(4) Pressure drops of air (inches H^20) and water (lb/in^2).
(5) Velocity of water at inlet connection (feet per second (ft/s)).
(6) Velocities of water through tubes and air through bundle (ft/s).
(7) Number of passes of air and water through cooler.
(8) Design pressure of waterside (lb/in^2).[1]
(9) Hydrostatic test pressure, waterside (lb/in^2).[2]
(10) Test pressure, air, leak-off system (lb/in^2).

[1] For class 1 coolers, no design pressure entry shall be made for the salt waterside.
[2] For class 1 coolers, in lieu of an entry in the tabulation opposite this heading, the space should be left blank. Beside the tabulation use an arrow leading in to indicate this blank space, and at the tail of the arrow add a note "FOR TUBE SIDE TEST PRESSURE SEE NOTE....ON.....DWG... ....NAVSEA NR....., 'TEST DATA,...'". This note is to be completed when the specific information becomes available.

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(11) Logarithmic mean temperature difference (°F).
(12) Heat transfer rates for both service and clean tube conditions (British thermal units (Btu) per hour per square foot outside surface per °F logarithmic mean temperature difference).
(13) Cooling surface square feet (ft^2). (For coolers using extended surface tubes, both the plain tube surface and the total surface including fins shall be shown.)
(14) Heat transfer capacity at design point (Btu per hour).
(15) Dry weight and operating weight of cooler (lb).

(c) A view of the identification plate with all data entered except date of manufacture, serial number, and the tube side maximum test pressure for class 1 air cooler applications. For this application use an arrow leading in, to indicate the blank spaces where tube side maximum test pressure would have been entered, with the notation "TUBE SIDE MAXIMUM TEST PRESSURE NOT TO BE STAMPED ON IDENTIFICATION PLATE" placed at the tail of the arrow. This view shall be enlarged when necessary to ensure that data entries will meet lettering height requirements (1/8 inch).
(d) The technical equipment manual number, provided such manuals are required by the order and the manufacturer has received the assigned number.
(e) The component identification number allowance parts list number (APL), if available.
(f) Certifying signature of manufacturer's responsible engineering representative. (This shall not be at lower than professional engineer level.)

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APPENDIX D

MANUAL TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that shall be included in technical manuals when required by the contract or order. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. TECHNICAL MANUAL

30.1 Technical manual content requirements. When required by the contract or order, bolt torquing procedures for waterbox and tube sheet bolting shall be fully described. Technical manuals shall include sufficient drawings to permit ship and shore activities to repair and maintain the unit without contractor's assistance. Information covering the air cooler shall be incorporated in the technical manual required for the parent machine, whenever feasible, and shall be in sufficient detail to meet all requirements for that manual. When such incorporation of cooler information is not feasible, a separate technical manual shall be furnished for the air cooler.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-19836B(SH)	COOLERS, FLUID, INDUSTRIAL, AIR, MOTOR AND GENERATOR, NAVAL SHIPBOARD

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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